

10 Vectors	10.1	Use vectors in two dimensions and in three dimensions	Students should be familiar with column vectors and with the use of $\mathbf{i}$ and $\mathbf{j}$ unit vectors in two dimensions and $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ unit vectors in three dimensions.
	10.2	Calculate the magnitude and direction of a vector and convert between component form and magnitude/direction form.	Students should be able to find a unit vector in the direction of $\mathbf{a}$, and be familiar with the notation $\mathbf{a}$.
	10.3	Add vectors diagrammatically and perform the algebraic operations of vector addition and multiplication by scalars, and understand their geometrical interpretations.	The triangle and parallelogram laws of addition. Parallel vectors.
	10.4	Understand and use position vectors; calculate the distance between two points represented by position vectors.	$\overrightarrow{OB} - \overrightarrow{OA} = \overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ The distance d between two points (x_1, y_1) and (x_2, y_2) is given by $d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$ In three dimensions, the distance d between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by $d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2$

Topics	What students need to learn:		
	Content		Guidance
10 Vectors <i>continued</i>	10.5	Use vectors to solve problems in pure mathematics and in context (including forces).	For example, finding position vector of the fourth corner of a shape (e.g. parallelogram) $ABCD$ with three given position vectors for the corners A, B and C. Contexts such as velocity, displacement, kinematics and forces will be covered in Paper 3, Sections 6.1, 7.3 and 8.1 – 8.4

Assessment information

- First assessment: May/June 2018.
- The assessments are 2 hours each.
- The assessments are out of 100 marks.
- Students must answer all questions.
- Calculators can be used in the assessments.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

Synoptic assessment

Synoptic assessment requires students to work across different parts of a qualification and to show their accumulated knowledge and understanding of a topic or subject area.

Synoptic assessment enables students to show their ability to combine their skills, knowledge and understanding with breadth and depth of the subject.

These papers assess synopticity.

Sample assessment materials

A sample paper and mark scheme for these papers can be found in the *Pearson Edexcel Level 3 Advanced GCE in Mathematics Sample Assessment Materials (SAMs)* document.